Checkpoint 1: Our Job Choice, Skill Gaps and Team Plan

Big Data Project Work | Team 5 | Due Tuesday 27 October 2026

Team	Team 5
Members	Boomikaa P, Chahatleen Kaur, Saba Gul, Vanshika Malviya, K. Shreya, Tushar Rathor
Job we chose	Data Engineer (Data Modeling) at IBM Consulting
Location	Hyderabad, Telangana, India (hybrid, full-time)
Link to the job	https://www.linkedin.com/jobs/view/4467603716/
Date we saved it	26 October 2026 (screenshot and text attached)

1. The job in simple words

IBM wants a data engineer who plans how a company's data should be organised. The person turns what the business needs into a clear design, first as a drawing (the logical model) and then as real tables in a system (the physical model). The person also makes sure the data stays correct and matches across different systems.

2. Why we chose this job

The skills in this job match what we can build in this project: organising data, cleaning it, keeping one correct record for each customer, and showing results in a dashboard. IBM is a well-known company, and the job lists must-have skills and nice-to-have skills clearly, which makes it easy to measure our gaps. Our team also has business and consulting experience, which helps with understanding what a business needs.

3. Words explained

Term	Simple meaning
Data model	A plan showing how data is organised and how the pieces connect.
ETL / ELT	Collecting data, cleaning it and loading it where it will be used.
Master data management	Keeping one correct record for each customer or product, even if it appears many times.
Star layout	A simple way to arrange data so reports run easily: one main table in the middle, with smaller tables around it.
Bronze, silver, gold	Three steps of data quality: raw data, cleaned data, and final ready-to-use data.
Databricks	The free online platform we will use to build the project.

4. Our skill gaps

Each member rated themselves from 0 to 3: 0 = never heard of it, 1 = know the idea, 2 = used it with help, 3 = can do it alone. The gap score is: importance x (3 minus the team average). Importance is 2 for a must-have skill and 1 for a nice-to-have skill. A bigger score means a bigger gap that matters more to the employer.

A = Boomikaa P, B = Chahatleen Kaur, C = Saba Gul, D = Vanshika Malviya, E = K. Shreya, F = Tushar Rathor.

Skill in the job ad	Type	A	B	C	D	E	F	Avg	Gap
Using a data-design tool such as ERWin *	Must	0	0	0	0	0	0	0.00	**6.00**
Keeping one clean record per customer or product (master data management) *	Must	0	0	0	0	0	0	0.00	**6.00**
Designing how data is organised, on paper and in real tables *	Must	0	0	1	0	0	0	0.17	**5.67**
Moving and cleaning data automatically (ETL/ELT)	Must	1	1	1	0	0	1	0.67	**4.67**
Star-layout design for reports (dimensional modelling)	Nice	0	0	0	1	0	0	0.17	**2.83**
Tools for very large data (Hadoop, NoSQL)	Nice	0	0	1	1	0	0	0.33	**2.67**
Keeping data correct and consistent	Must	1	2	2	2	2	1	1.67	**2.67**
Understanding what the business needs	Must	1	2	3	1	2	1	1.67	**2.67**
Data warehouse basics	Nice	0	1	1	0	1	1	0.67	**2.33**
Data rules and quality checks	Nice	1	2	1	1	1	1	1.17	**1.83**

* These are our top 3 gaps.

5. Our top 3 gaps and how we will close them

#	Gap	How our project will close it
1	Using a data-design tool such as ERWin (score 6.0)	ERWin is paid software, so we will learn the same skill with a free drawing tool (for example draw.io). We will draw the logical model and write the table definitions (physical model).
2	One clean record per customer or product (score 6.0)	We will remove duplicates and match records so each customer or product appears once: one agreed description per product code, and one customer table with one row per customer number.
3	Designing how data is organised (score 5.7)	We will first draw the design, then build the real tables in a star layout: one sales table in the middle with customer, product, date and country tables around it.

6. Our team plan (agile charter)

Our goal	Show that we can do IBM's Data Engineer job by building a well-organised data project on Databricks. It ends with a dashboard that helps a business make a decision.
What we will do	Load raw data, clean it, keep one record per customer or product, build the star-layout tables, check data quality, and make a dashboard and a guide for others to rerun our work.
What we will not do	Use paid tools, process live streaming data, copy data from websites, or contact or apply to the company.
We are done when	The work runs from start to finish, a second person has checked it, and every member can explain each step.

Who does what (we swap roles after Checkpoint 3)

Name	Role	What they do
Boomikaa P	Project Captain	Keeps track of deadlines and submissions
Chahatleen Kaur	Data Wrangler	Loads and cleans the data
Tushar Rathor	Data Wrangler (shared)	Shares the loading and cleaning work with Chahatleen, starting with the Bronze load
Saba Gul	Analyst and Governance Guardian	Builds final tables and finds results; checks licence, privacy and data quality; keeps the learning log
Vanshika Malviya	Storyteller	Makes the dashboard and presentation
K. Shreya	Quality Reviewer	Second-checks the work from start to finish, reruns the guide to test it, and helps the team prepare for questions

The sixth member shares a role, as the project guide allows.

Timeline

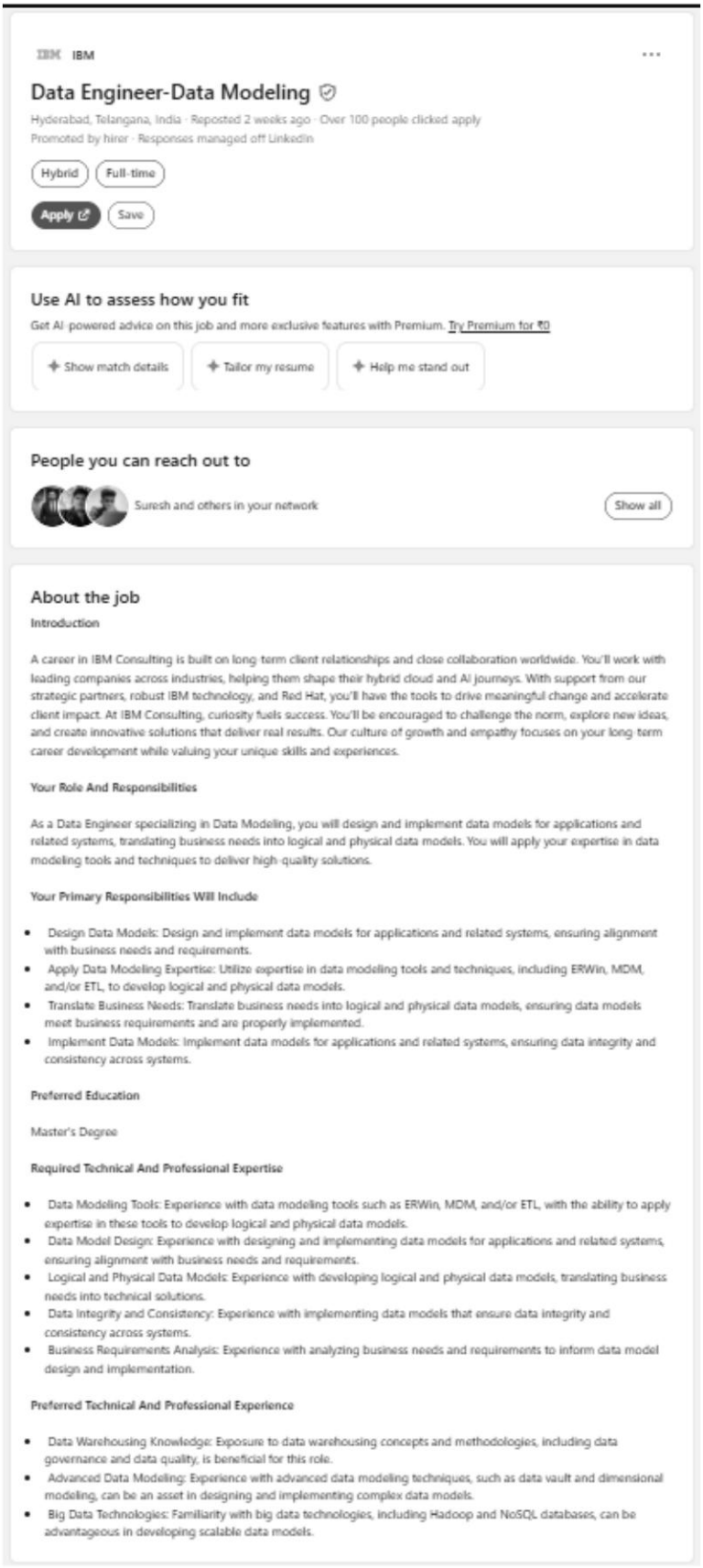
When	What we do	Due
20 to 27 Oct	Choose the job, rate our skills, find the top 3 gaps	27 Oct
27 Oct to 3 Nov	Choose the data and write a one-page project plan	3 Nov
3 to 17 Nov	Load raw data (Bronze) and clean it (Silver)	17 Nov
17 to 24 Nov	Build final tables and the dashboard; write the guide	24 Nov
24 to 27 Nov	Write the final report; book the presentation slot	27 Nov
30 Nov to 2 Dec	Present: demo, questions and mock interview	Presentation

How we work together

We meet for 15 minutes every week. We keep a shared log of what went wrong and what we learned. We rate each other's contribution at the end. If we use an AI tool, we write it down. Everyone runs the final work once on their own account.

Appendix A: Screenshot of the live job posting

Taken on 26 October 2026 from https://www.linkedin.com/jobs/view/4467603716/



IBM IBM

Data Engineer-Data Modeling

Hyderabad, Telangana, India · Reposted 2 weeks ago · Over 100 people clicked apply

Promoted by hirer · Responses managed off LinkedIn

Hybrid | Full-time

Apply | Save

Use AI to assess how you fit

Get AI-powered advice on this job and more exclusive features with Premium. Try Premium for ₹0

Show match details | Tailor my resume | Help me stand out

People you can reach out to

Suresh and others in your network

Show all

About the job

Introduction

A career in IBM Consulting is built on long-term client relationships and close collaboration worldwide. You'll work with leading companies across industries, helping them shape their hybrid cloud and AI journeys. With support from our strategic partners, robust IBM technology, and Red Hat, you'll have the tools to drive meaningful change and accelerate client impact. At IBM Consulting, curiosity fuels success. You'll be encouraged to challenge the norm, explore new ideas, and create innovative solutions that deliver real results. Our culture of growth and empathy focuses on your long-term career development while valuing your unique skills and experiences.

Your Role And Responsibilities

As a Data Engineer specializing in Data Modeling, you will design and implement data models for applications and related systems, translating business needs into logical and physical data models. You will apply your expertise in data modeling tools and techniques to deliver high-quality solutions.

Your Primary Responsibilities Will Include

- Design Data Models: Design and implement data models for applications and related systems, ensuring alignment with business needs and requirements.
- Apply Data Modeling Expertise: Utilize expertise in data modeling tools and techniques, including ERWin, MDM, and/or ETL, to develop logical and physical data models.
- Translate Business Needs: Translate business needs into logical and physical data models, ensuring data models meet business requirements and are properly implemented.
- Implement Data Models: Implement data models for applications and related systems, ensuring data integrity and consistency across systems.

Preferred Education

Master's Degree

Required Technical And Professional Expertise

- Data Modeling Tools: Experience with data modeling tools such as ERWin, MDM, and/or ETL, with the ability to apply expertise in these tools to develop logical and physical data models.
- Data Model Design: Experience with designing and implementing data models for applications and related systems, ensuring alignment with business needs and requirements.
- Logical and Physical Data Models: Experience with developing logical and physical data models, translating business needs into technical solutions.
- Data Integrity and Consistency: Experience with implementing data models that ensure data integrity and consistency across systems.
- Business Requirements Analysis: Experience with analyzing business needs and requirements to inform data model design and implementation.

Preferred Technical And Professional Experience

- Data Warehousing Knowledge: Exposure to data warehousing concepts and methodologies, including data governance and data quality, is beneficial for this role.
- Advanced Data Modeling: Experience with advanced data modeling techniques, such as data vault and dimensional modeling, can be an asset in designing and implementing complex data models.
- Big Data Technologies: Familiarity with big data technologies, including Hadoop and NoSQL databases, can be advantageous in developing scalable data models.

Appendix B: Saved copy of the job posting text (original wording)

Source: https://www.linkedin.com/jobs/view/4467603716/ | IBM Consulting, Data Engineer (Data Modeling) | Copied by hand on 26 October 2026.

Introduction

A career in IBM Consulting is built on long-term client relationships and close collaboration worldwide. You'll work with leading companies across industries, helping them shape their hybrid cloud and AI journeys. With support from our strategic partners, robust IBM technology, and Red Hat, you'll have the tools to drive meaningful change and accelerate client impact. At IBM Consulting, curiosity fuels success. You'll be encouraged to challenge the norm, explore new ideas, and create innovative solutions that deliver real results. Our culture of growth and empathy focuses on your long-term career development while valuing your unique skills and experiences.

Your Role and Responsibilities

As a Data Engineer specializing in Data Modeling, you will design and implement data models for applications and related systems, translating business needs into logical and physical data models. You will apply your expertise in data modeling tools and techniques to deliver high-quality solutions.

Your primary responsibilities will include

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Translate Business Needs: Translate business needs into logical and physical data models, ensuring data models meet business requirements and are properly implemented.

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Data Model Design: Experience with designing and implementing data models for applications and related systems, ensuring alignment with business needs and requirements.

Logical and Physical Data Models: Experience with developing logical and physical data models, translating business needs into technical solutions.

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Data Warehousing Knowledge: Exposure to data warehousing concepts and methodologies, including data governance and data quality, is beneficial for this role.

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Big Data Technologies: Familiarity with big data technologies, including Hadoop and NoSQL databases, can be advantageous in developing scalable data models.